First Isomorphism Theorem for Groups

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of first isomorphism theorem for groups, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-faculty node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

MA-02 / Modern Algebra / Report v1

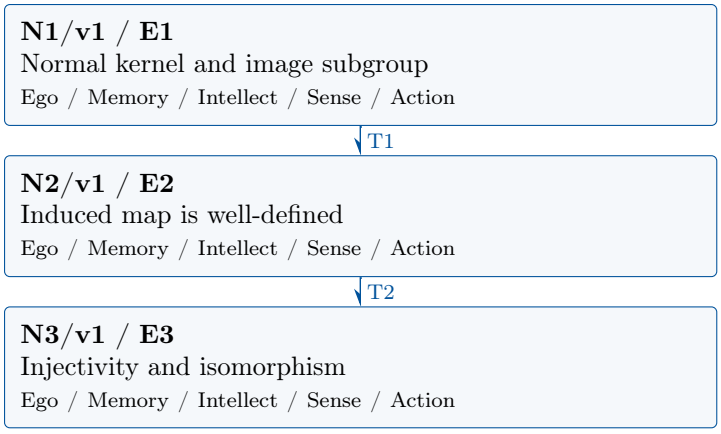
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For a group homomorphism $\varphi : G \to K$, prove* $\ker \varphi$ *is normal and* $G/\ker \varphi \cong \operatorname{im} \varphi$.

A. Foundations and permitted imports

Groups need not be finite. A homomorphism satisfies $\varphi(gh) = \varphi(g)\varphi(h)$. For a normal subgroup N the quotient has *multiplication* $(gN)(hN) = ghN$. The proof audits normality and representative independence before using that construction.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five faculties. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. A normal kernel N and subgroupim φ.

Receiving activity. N2: Induced map is well-defined

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: NormalN; proposed $\Phi(gN) = \varphi(g)$; prior exports E1/v1

T2: N2/E2/v1 to N3. A well-defined surjective homomorphism $\Phi : G/N \to \operatorname{im}\varphi$.

Receiving activity. N3: Injectivity and isomorphism

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Induced Φ; prior exports E2/v1

III. AGENT ROLES AND FIVE-FACULTY ACTIVITIES

A. N1/v1: Normal kernel and image subgroup

Object and role. Use inverse and conjugation identities.

Inputs. $\varphi : G \to K$ a homomorphism; foundational imports stated in §1

E — Ego

Own E1/v1: A normal kernel N and subgroupim φ.

M — Memory

Retain $\varphi : G \to K$ a homomorphism; foundational imports stated in §1; keep their assumptions and versions attached.

I — Intellect

Use inverse and conjugation identities.

S — Sense

The theorem's codomain conclusion is $\operatorname{im}\varphi$, not all of K.

A — Action

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 The kernel $N = \{g : \varphi(g) = 1\}$ contains 1 and is closed under products and inverses.

N1.2 For $n \in N$ and $g \in G$, $\varphi(gng^{-1}) = \varphi(g)1\varphi(g)^{-1} = 1$, so N is normal.

N1.3 The image contains 1 and is closed under products and inverses, hence is a subgroup of K.

N1.4 Quotient multiplication is valid because N is normal.

Export E1/v1. A normal kernel N and subgroupim φ.

Acceptance gate. The theorem's codomain conclusion is $\operatorname{im}\varphi$, not all of K. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: The theorem's codomain conclusion is $\operatorname{im}\varphi$, not all of K. Recheck dependent nodes N2, N3.

B. N2/v1: Induced map is well-defined

Object and role. Check representative equality before algebra.

Inputs. NormalN; proposed $\Phi(gN) = \varphi(g)$; prior exports E1/v1

E — Ego

Own E2/v1: A well-defined surjective homomorphism $\Phi : G/N \to \operatorname{im}\varphi$.

M — Memory

Retain NormalN; proposed $\Phi(gN) = \varphi(g)$; prior exports E1/v1; keep their assumptions and versions attached.

I — Intellect

Check representative equality before algebra.

S — Sense

A formula on cosets is not automatically well-defined.

A — Action

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 If $gN = hN$ then $h^{-1}g \in N$, so $\varphi(h)^{-1}\varphi(g) = 1$ and $\varphi(g) = \varphi(h)$.

N2.2 Thus Φ is independent of representatives.

N2.3 It preserves multiplication since $\Phi(ghN) = \varphi(gh) = \varphi(g)\varphi(h) = \Phi(gN)\Phi(hN)$.

N2.4 It is ontoim φ by the definition of image.

Export E2/v1. A well-defined surjective homomorphism $\Phi : G/N \to \operatorname{im}\varphi$.

Acceptance gate. A formula on cosets is not automatically well-defined. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: A formula on cosets is not automatically well-defined. Recheck dependent nodes N3.

C. N3/v1: Injectivity and isomorphism

Object and role. Use equality of images to recover equality of cosets.

Inputs. Induced Φ; prior exports E2/v1

E — Ego

Own E3/v1: The first isomorphism theorem for arbitrary groups.

M — Memory

Retain Induced Φ; prior exports E2/v1; keep their assumptions and versions attached.

I — Intellect

Use equality of images to recover equality of cosets.

S — Sense

Equal finite sizes cannot substitute for injectivity in infinite groups.

A — Action

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 If $\Phi(gN) = \Phi(hN)$, then $\varphi(h^{-1}g) = 1$, so $h^{-1}g \in N$ and $gN = hN$.

N3.2 Hence Φ is injective as well as surjective and a homomorphism.

N3.3 Its inverse preserves products because preimages are unique, so it is a group isomorphism.

N3.4 This proves $G/\ker\varphi \cong \operatorname{im}\varphi$ without any cardinality argument.

Export E3/v1. The first isomorphism theorem for arbitrary groups.

Acceptance gate. Equal finite sizes cannot substitute for injectivity in infinite groups. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: Equal finite sizes cannot substitute for injectivity in infinite groups. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Normal kernel and image subgroup

1. The kernel $N = \{g : \varphi(g) = 1\}$ contains 1 and is closed under products and inverses.
2. For $n \in N$ and $g \in G$, $\varphi(gng^{-1}) = \varphi(g)1\varphi(g)^{-1} = 1$, so N is normal.
3. The image contains 1 and is closed under products and inverses, hence is a subgroup of K.
4. Quotient multiplication is valid because N is normal.

B. §4.N2 — Induced map is well-defined

Define $\Phi : G/N \to \operatorname{im}\varphi$ by $\Phi(gN) = \varphi(g)$.

1. If $gN = hN$ then $h^{-1}g \in N$, so $\varphi(h)^{-1}\varphi(g) = 1$ and $\varphi(g) = \varphi(h)$.
2. Thus Φ is independent of representatives.
3. It preserves multiplication since $\Phi(ghN) = \varphi(gh) = \varphi(g)\varphi(h) = \Phi(gN)\Phi(hN)$.
4. It is ontoim φ by the definition of image.

C. §4.N3 — Injectivity and isomorphism

1. If $\Phi(gN) = \Phi(hN)$, then $\varphi(h^{-1}g) = 1$, so $h^{-1}g \in N$ and $gN = hN$.
2. Hence Φ is injective as well as surjective and a homomorphism.
3. Its inverse preserves products because preimages are unique, so it is a group isomorphism.
4. This proves $G/\ker\varphi \cong \operatorname{im}\varphi$ without any cardinality argument.

□

V. WORKED ORCHESTRATION

For $\varphi : Z \to Z/6Z$ sending an integer to its residue, $\ker\varphi = 6Z$ andim $\varphi = Z/6Z$. The induced map takes the additive cosetn+6Z to[n], giving the explicit isomorphism. For $\varphi(n) = 2n$ intoZ, the image is 2Z, not the whole codomain.

A. Boundary and hypothesis audit

For the $inclusion 2Z \hookrightarrow Z$ the kernel is 0; the quotient is 2Z and the theorem identifies it with its image 2Z. Sense rejects replacing the image by Z through the induced inclusion, since that map is not onto.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Ego → Memory:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Intellect → Action:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Sense → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Memory + Intellect:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Action:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Ego:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

For the $inclusion 2Z \hookrightarrow Z$ the kernel is 0; the quotient is 2Z and the theorem identifies it with its image 2Z. Sense rejects replacing the image by Z through the induced inclusion, since that map is not onto.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.